Modularity Matrix Approach for Graph Learning?

Parts I and II: what modularity adds to Laplacian graph learning on stock returns

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Abstract. The proposal asks whether Newman's modularity matrix, the standard tool for finding communities in a network, can improve the way Cardoso and Palomar (2020) learn a network of stocks from their returns. This report answers its first two parts. *Part I*: under the paper's rule that every stock has total link weight 1, the modularity matrix is an exact mirror image of the graph Laplacian, so modularity adds nothing new; I prove the three claims of the proposal, show where the needed condition fails, and confirm the mirror image on the paper's own learned graph, where it holds to within the solver's tolerance. *Part II*: the Laplacian learner already removes the market's common movement on its own, exactly when all stocks have the same beta. On the paper's S&P data this built-in removal is enough, and four standard ways of removing the market explicitly all make the result worse. In simulated markets where betas vary within sectors, the built-in removal fails suddenly once beta inequality passes a threshold, and explicit removal then fixes it. A two-step perturbation bound is proved; its first step is realistic, its second is too loose to be useful here, and the measured switch point is reported instead.

Code: `modularity_graph_learning/`. Every number is printed by `report_numbers.py`; every figure is drawn from the learned graphs by `report_figures.py`.

Contents

1 Introduction **1**

1.1 The starting point: learning a network of stocks . . . 1
1.2 What the proposal asks . . . 1
1.3 Findings at a glance . . . 1
1.4 How to read this report . . . 2

2 Background: the objects and the terms **2**

2.1 The data . . . 2
2.2 Graphs, weights and degrees . . . 2
2.3 The graph Laplacian . . . 3
2.4 What the eigenvalues of a Laplacian say . . . 3
2.5 Modularity and the modularity matrix . . . 5
2.6 The learners of Paper 1 . . . 6

3 Part I: modularity under the unit-degree rule **6**

3.1 The question . . . 6
3.2 The reflection lemma and the three claims . . . 7
3.3 Why (b) needs $\lambda_k \leq 1$ . . . 7
3.4 When do the Laplacian and modularity agree in general? . . . 8
3.5 The check on Paper 1's learned graph . . . 9
3.6 Loosening the degree rule . . . 9
3.7 Where claim (c) applies . . . 13
3.8 Part I conclusions . . . 13

4 Part II: the market as a built-in null model **13**

4.1 The question . . . 13
4.2 The learner removes the market on its own . . . 13
4.3 Real data: the S&P panel . . . 14
4.4 Simulated markets: when betas cut across sectors . . . 17
4.5 A perturbation bound . . . 18
4.6 The bound against the measurements . . . 21
4.7 Part II conclusions . . . 22

5 Status, deviations and next steps **23**

5.1 Status of every claim . . . 23
5.2 Deviations . . . 23
5.3 Suggested next steps . . . 24
5.4 Reproducing . . . 24

A Glossary **25**

References **25**

1 Introduction

1.1 The starting point: learning a network of stocks

Cardoso and Palomar (2020), *Learning undirected graphs in financial markets* (arXiv:2005.09958; "Paper 1" below), turn a table of daily stock returns into a network. Each stock is a node; a link between two stocks carries a weight that says how directly their returns are tied together once all the other stocks are accounted for. The learned network is then read for structure: which groups of stocks hang together (clusters), and how tightly the market as a whole is connected (used as a stress indicator). Figure 1 shows the pipeline used throughout this report.

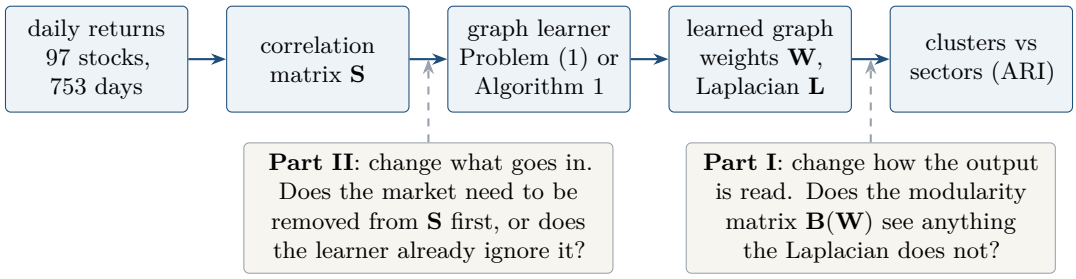


Figure 1: The pipeline of Paper 1 as used in this report, and where the two parts of the proposal act on it.

1.2 What the proposal asks

The proposal, *Modularity Matrix Approach for Graph Learning?*, starts from an observation: the graph Laplacian that Paper 1 learns, and the modularity matrix that network science uses to find communities, are built from the same weights. It asks, in three parts, whether modularity can add something.

- **Part I** asks what modularity says about the graphs Paper 1 already learns. Paper 1's clustering method forces every stock to have total link weight 1; under that rule the proposal conjectures that the two matrices are mirror images and lists three consequences, labelled (a), (b) and (c).
- **Part II** asks about the market. Modularity methods subtract an explicit model of "what everything has in common" (a *null model*) before looking for groups. Paper 1 never does. The proposal conjectures that the Laplacian removes the market on its own, and asks when that is enough, given that stocks react to the market by different amounts (their *betas*).
- **Part III** proposes a new learner with a modularity term in place of the unit-degree rule. It is not covered here: as written its objective is unbounded below (reported in the 18 October note), and it waits on a choice of repair.

1.3 Findings at a glance

Part I: under the unit-degree rule, modularity adds nothing new

- The modularity matrix is the Laplacian reflected: $\mathbf{B} = \mathbf{I} - \mathbf{L} - \mathbf{1}\mathbf{1}^\top/p$, so every eigenvalue λ of $\mathbf{L}$ becomes $1-\lambda$ of $\mathbf{B}$ with the same eigenvector. All three claims follow; they are proved in Section 3.2.
- Claim (b) needs a condition ($\lambda_k \leq 1$). I give its exact form without the condition and two small graphs where it fails (Section 3.3).
- On Paper 1's own learned graph every eigenvalue pair matches the mirror image to within 0.006, and the modularity matrix finds exactly the three sectors (Section 3.5).
- Without the unit-degree rule the algebra stops holding, but the clusters do not change; a degree-normalised version of the identity holds for any degrees (Sections 3.4 and 3.6).

Part II: the market is removed for free, unless betas cut across groups

- The learner only ever sees returns with each day's average removed. Equal betas are therefore removed exactly; unequal betas leave a penalty on links between high-beta and low-beta stocks (Section 4.2).
- On the S&P data, Algorithm 1 recovers the sectors exactly, and every explicit market removal does worse (ARI 0.830 to 0.896 against 1.000), because here betas follow sectors (Section 4.3).
- In simulated markets with high and low betas inside every sector, the clusters switch from sectors to beta groups over a narrow range of beta inequality; explicit removal recovers the sectors every time (Section 4.4).
- A two-step perturbation bound is proved. Step 1 is within a factor 1.65 to 4.84 of the truth; Step 2 (Davis-Kahan) is too loose to be informative. The switch happens when the change in the graph reaches the size of its eigengap (Section 4.6).

1.4 How to read this report

Section 2 introduces every object and term used later, each with a small worked picture; readers who know spectral graph theory can skip to Section 3. Terms are also collected in the glossary (Appendix A). Each result carries one of three labels: *reproduced* (the claim holds in our computations), *partial* (it holds with qualifications that are stated), or *not reproduced / not supported*. Negative results are reported as prominently as positive ones, and every deviation from the paper or the proposal is listed in Section 5. The suggested next steps are suggestions; which to pursue is the professor's decision.

2 Background: the objects and the terms

2.1 The data

Paper 1's clustering experiment uses $p = 97$ stocks from the S&P 500 in three sectors: 52 Industrials, 28 Consumer Staples and 17 Energy. For each stock i and trading day t the *log return* $x_{t,i} = \log(P_{t,i}/P_{t-1,i})$ is the day's relative price change. We use the paper's window of $T = 753$ trading days, 5 January 2016 to 31 December 2018.

Correlation matrix S
The $p \times p$ matrix whose entry S_{ij} is the sample correlation between the daily returns of stocks i and j over the T days. $S_{ii} = 1$; values near 1 mean the two stocks rise and fall together. $\mathbf{S}$ is the only input the learners see.

Figure 2 shows the two features that drive this report. The sectors are visible, which is what a good learner should recover. And everything is positively correlated, which is the market factor that Part II is about.

2.2 Graphs, weights and degrees

Weighted graph
A set of p nodes (here, stocks) and a symmetric weight matrix $\mathbf{W}$ with $W_{ij} = W_{ji} \geq 0$ and $W_{ii} = 0$. $W_{ij} > 0$ means there is a link (an *edge*) between i and j, and its size is the link's strength.

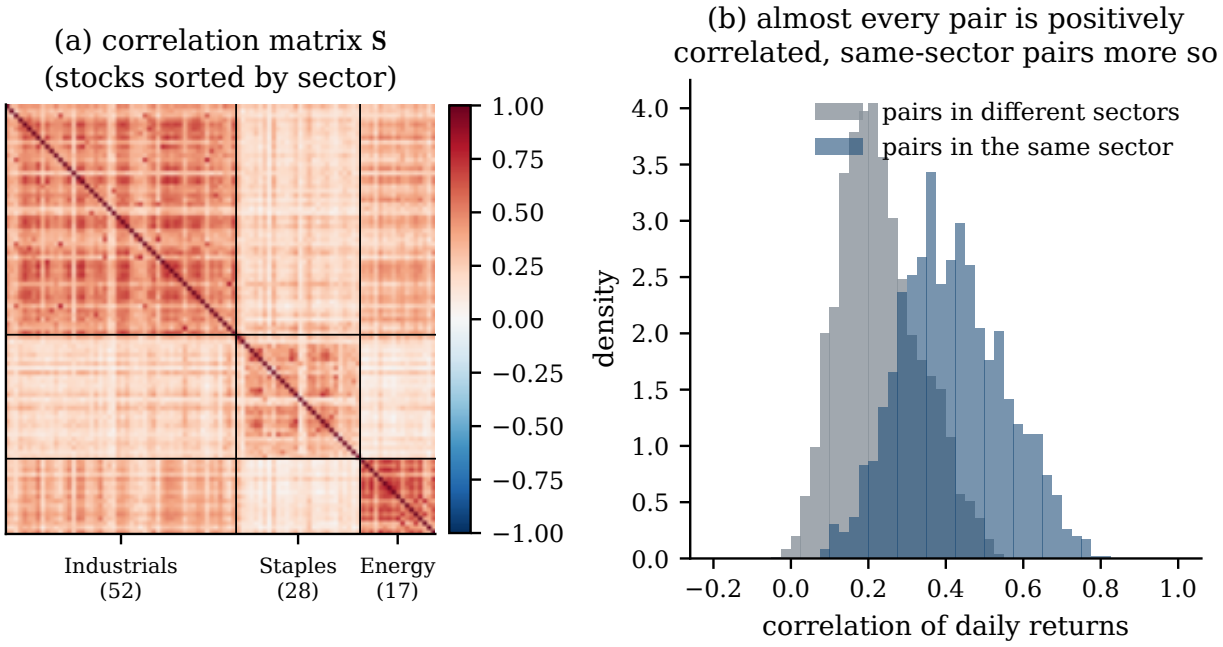


Figure 2: The input. (a) The correlation matrix with stocks sorted by sector: the three darker diagonal blocks are the sectors, but nearly every entry is positive, because all stocks share the market's daily ups and downs. (b) Same-sector pairs are more correlated than cross-sector pairs, but the two distributions overlap heavily.

Degree d_i
The total weight of node i's links, $d_i = \sum_j W_{ij}$. Collected in the vector $\mathbf{d} = \mathbf{W1}$, where $\mathbf{1}$ is the vector of all ones, and in the diagonal matrix $\mathbf{D} = \mathrm{Diag}(\mathbf{d})$. The total weight of the graph is $2m = \sum_i d_i = \mathbf{1}^\top \mathbf{d}$ (each edge counted from both ends).

A graph is *regular* if every degree is the same number c, and has *unit degree* if $c = 1$. A *connected component* is a group of nodes joined to each other by paths of edges but with no edge to any node outside the group.

2.3 The graph Laplacian

Graph Laplacian L
$\mathbf{L} = \mathbf{D} - \mathbf{W}$: the degrees on the diagonal and minus the weights off it. For any vector $\mathbf{x}$ with one value per node,

$$\mathbf{x}^\top \mathbf{L}\mathbf{x} = \sum_{i<j} W_{ij}(x_i - x_j)^2. \tag{1}$$

Equation (1) is the key to everything below. It says that $\mathbf{L}$ measures how much a node signal $\mathbf{x}$ *changes across edges*: each edge charges its weight times the squared difference of its two ends. A signal that is constant on tightly linked groups and changes only across weak links is cheap; a signal that alternates between neighbours is expensive (Figure 3). In particular $\mathbf{L1} = \mathbf{0}$: moving every node by the same amount costs nothing.

2.4 What the eigenvalues of a Laplacian say

$\mathbf{L}$ is symmetric and, by (1), positive semidefinite, so it has real eigenvalues $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_p$ with orthonormal eigenvectors. Each eigenvector is a node signal, and its eigenvalue is its cost (1). Three facts are used repeatedly.

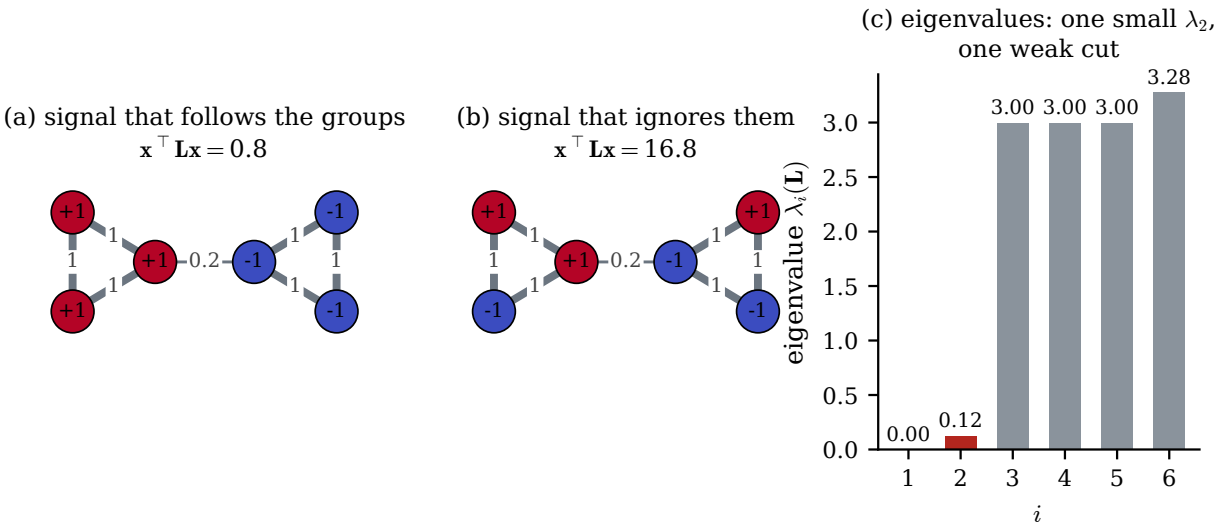


Figure 3: The Laplacian on a toy graph: two triangles (edge weights 1) joined by a weak edge (weight 0.2). (a) A signal that is $+1$ on one triangle and -1 on the other only changes across the weak edge, so $\mathbf{x}^\top \mathbf{L}\mathbf{x}$ is small. (b) A signal that alternates between neighbours is expensive. (c) The eigenvalues of $\mathbf{L}$: 0 (the constant signal), then one small eigenvalue $\lambda_2 = 0.12$ whose eigenvector is the signal in (a), then a jump.

- **Zero eigenvalues count components.** A signal that is constant on each component costs nothing, so a graph with k components has exactly k zero eigenvalues, that is, $\text{rank}(\mathbf{L}) = p - k$ (Figure 4a).
- **Small eigenvalues mark weak cuts.** If the components are joined by weak links, the zeros become small positive eigenvalues, still well separated from the rest (Figure 4b). λ_2 is called the *algebraic connectivity*: it is 0 exactly when the graph is disconnected, and small when the graph is close to splitting. Its eigenvector is the *Fiedler vector*.
- **The eigengap measures how clear the groups are.** The gap $\lambda_{k+1} - \lambda_k$ between the k small eigenvalues and the rest says how sharply the graph separates into k groups. It reappears in Part II, where it decides how much perturbation the groups can survive.

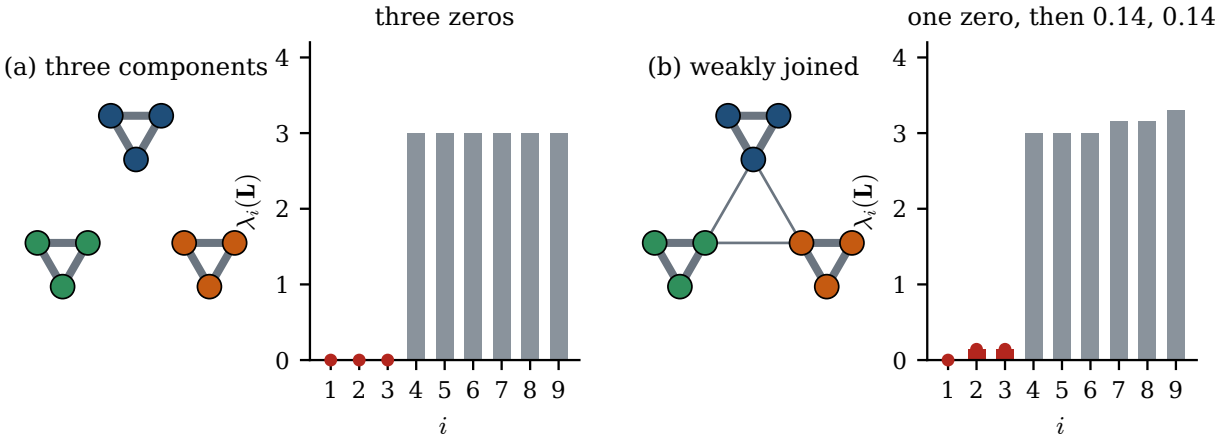


Figure 4: (a) Three separate triangles: three zero eigenvalues (red), one per component. (b) The same triangles joined by weak edges (weight 0.15): one zero, then two small eigenvalues, then a gap. Counting small eigenvalues counts groups.

Spectral clustering, k-means and ARI

Spectral clustering gives each node the coordinates it has in the k eigenvectors with the smallest eigenvalues, then groups the nodes with *k-means*, which splits points into k groups so that each point is close to its group's average. For a graph with k components this simply returns the components. The *adjusted Rand index* (ARI) compares two ways of grouping the same items: 1 means identical groupings, about 0 means no more agreement than chance, and negative values mean less. Throughout, ARI compares learned clusters with the sector labels.

2.5 Modularity and the modularity matrix

Network science finds communities differently. Newman's *modularity* asks whether a group has more internal weight than one would expect by chance, and so needs a model of "chance": a *null model*.

Null model and modularity matrix B

The standard (configuration) null model keeps every node's degree and wires the weight at random, so the expected weight between i and j is $d_i d_j / 2m$. The *modularity matrix* is observed minus expected,

$$\mathbf{B} = \mathbf{W} - \frac{\mathbf{d}\mathbf{d}^\top}{2m}, \qquad B_{ij} = W_{ij} - \frac{d_i d_j}{2m}.$$

For a split of the nodes into groups, the *modularity* Q adds up B_{ij} over pairs in the same group (divided by $2m$): positive Q means the groups hold more weight than chance.

Maximising Q exactly is hard, so it is relaxed: replacing group labels by real-valued vectors, the best two-way split is read from the top eigenvector of $\mathbf{B}$, and the best k-way split from the top $k-1$ eigenvectors, through the *relaxed modularity* $\max \operatorname{tr}(\mathbf{U}^\top \mathbf{B}\mathbf{U})$ over $p \times (k-1)$ matrices $\mathbf{U}$ with orthonormal columns. Write the eigenvalues of $\mathbf{B}$ in decreasing order, $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_p$. Since $\mathbf{B}\mathbf{1} = \mathbf{0}$, 0 is always one of them. Newman's *divisibility test* says a network is worth splitting at all only if $\mu_1 > 0$ (Figure 5).

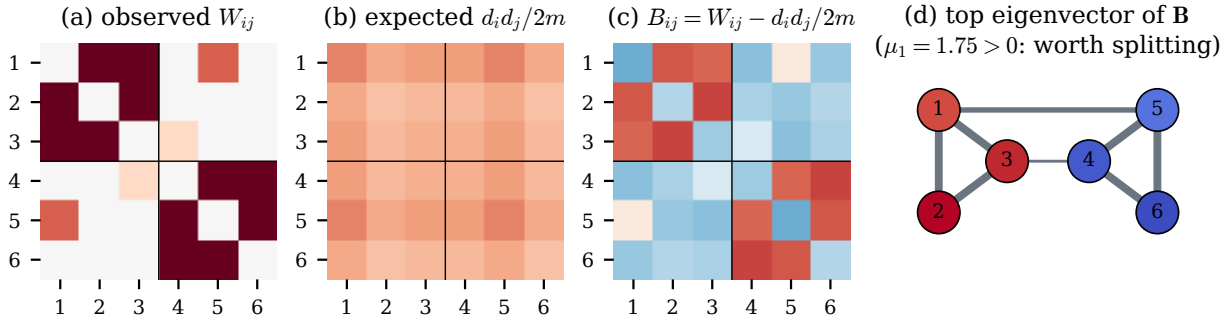


Figure 5: Modularity on the toy graph of Figure 3 with one extra cross edge (nodes 1 and 5). (a) Observed weights. (b) Weights expected by chance given the degrees. (c) Their difference $\mathbf{B}$: positive (red) inside the triangles, negative (blue) between them. (d) The top eigenvector of $\mathbf{B}$ separates the two triangles, and $\mu_1 > 0$ says splitting is worthwhile.

The parallel with Section 2.4 is the starting point of the proposal: the Laplacian finds groups through its *smallest* eigenvalues, modularity through its *largest*. Part I asks when these are the same thing.

Normalised versions

Dividing by the degrees gives the *normalised Laplacian* $\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$ and the *normalised modularity matrix* $\mathbf{B}_n = \mathbf{D}^{-1/2}\mathbf{B}\mathbf{D}^{-1/2}$. They treat a weakly connected node and a strongly connected node on the same footing, and appear in Section 3.4.

2.6 The learners of Paper 1

Paper 1 models the returns as a *Gaussian Markov random field*: Gaussian with precision (inverse covariance) matrix $\mathbf{L}$. In such a model $L_{ij} = -W_{ij}$ is nonzero exactly when stocks i and j are linked after accounting for all the others. Fitting the model to $\mathbf{S}$ by maximum likelihood gives an objective with two parts, both in terms of $\mathbf{L}$.

Problem (1): the basic learner

$$\min_{\mathbf{L} \text{ a Laplacian}} \operatorname{tr}(\mathbf{LS}) - \log \operatorname{gdet}(\mathbf{L}) + \beta \sum_{i<j} W_{ij}.$$

$\operatorname{tr}(\mathbf{LS}) = \sum_{i<j} W_{ij}(S_{ii}+S_{jj}-2S_{ij})$ is the *data fit*: by (1) it is small when strong links join stocks whose returns move together. gdet, the *generalised determinant*, is the product of the nonzero eigenvalues; the $-\log \operatorname{gdet}$ term stops all weights collapsing to zero. The last term, with *sparsity weight* β, favours fewer links. We use $\beta = 10$, the setting of the paper's Fig. 1. Problem (1) is convex and returns one connected graph; clusters are then read by spectral clustering.

Algorithm 1: the k-component learner

Paper 1's Algorithm 1 adds two requirements: the graph must have exactly k components (here $k = 3$, one per sector), and every degree must be 1, $\operatorname{diag}(\mathbf{L}) = \mathbf{1}$, so that no stock can be cut off with zero links. The component requirement is imposed through *Ky Fan's theorem*: the sum of the k smallest eigenvalues of $\mathbf{L}$ equals $\min \operatorname{tr}(\mathbf{V}^\top \mathbf{L}\mathbf{V})$ over $p \times k$ matrices $\mathbf{V}$ with orthonormal columns, and adding $\eta \operatorname{tr}(\mathbf{V}^\top \mathbf{L}\mathbf{V})$ to the objective pushes those k eigenvalues to 0. The algorithm alternates between choosing $\mathbf{V}$ (the bottom k eigenvectors of the current $\mathbf{L}$) and re-solving for $\mathbf{L}$. Clusters are its connected components.

Three implementation choices are ours and recorded as deviations. The paper does not report the Fan weight; we use $\eta = 300$. The degree rule is imposed as a penalty $\frac{\rho}{2}\|\operatorname{diag}(\mathbf{L}) - \mathbf{1}\|^2$ with $\rho = 100$, so learned degrees are close to 1 but not exactly 1. And Algorithm 1 starts from the Problem (1) graph: from the paper's uniform starting graph the components are set by round-off error and do not match the sectors (18 October note).

3 Part I: modularity under the unit-degree rule

3.1 The question

Algorithm 1 forces every degree to be 1. With $\mathbf{d} = \mathbf{1}$ the null model's expected weight $d_i d_j / 2m$ becomes the same number, $1/p$, for every pair: chance predicts a uniform graph. So the modularity matrix no longer depends on the degrees at all, and it is natural to suspect it carries no information beyond $\mathbf{L}$. The proposal makes this precise in three claims. In words:

(a) **Counting clusters.** The graph has k components exactly when $\mathbf{B}$ has the eigenvalue 1 repeated $k - 1$ times. So modularity counts clusters the same way the Laplacian does.

(b) **Algorithm 1 is secretly maximising modularity.** The Ky Fan term that Algorithm 1 minimises equals a constant minus the relaxed $(k-1)$-way modularity, provided $\lambda_k \leq 1$.

(c) **The stress indicator is a modularity test.** The top modularity eigenvalue is $\mu_1 = \max\{0, 1 - \lambda_2\}$. Paper 1 tracks the algebraic connectivity λ_2 of rolling-window graphs as a market-stress indicator, and our replication of its trading rule invests only when $\lambda_2 < \tau = 1$. Claim (c) says that, under unit degree, this rule is exactly Newman's divisibility test $\mu_1 > 0$.

3.2 The reflection lemma and the three claims

When $\mathbf{d} = \mathbf{1}$: $\mathbf{D} = \mathbf{I}$, the total weight is $2m = p$, and so

$$\mathbf{B} = \mathbf{W} - \tfrac{1}{p}\mathbf{1}\mathbf{1}^\top = \mathbf{I} - \mathbf{L} - \tfrac{1}{p}\mathbf{1}\mathbf{1}^\top. \tag{2}$$

This is a property of the *learned* graph, not an assumption about the data. The eigenvalues of a unit-degree Laplacian lie in $[0, 2]$, since $\mathbf{W} = \mathbf{I} - \mathbf{L}$ has rows of absolute sum 1 and hence eigenvalues in $[-1, 1]$.

Lemma 1 (Reflection)**.** *Under* $\mathbf{d} = \mathbf{1}$*, take an orthonormal eigenbasis of* $\mathbf{L}$ *with* $\mathbf{v}_1 = \mathbf{1}/\sqrt{p}$ *and* $\mathbf{L}\mathbf{v}_i = \lambda_i \mathbf{v}_i$*. Then* $\mathbf{B}\mathbf{v}_1 = \mathbf{0}$ *and* $\mathbf{B}\mathbf{v}_i = (1 - \lambda_i)\mathbf{v}_i$ *for* $i \geq 2$*. So* $\mathbf{L}$ *and* $\mathbf{B}$ *have the same eigenvectors, and* $\operatorname{spec}(\mathbf{B}) = \{0\} \cup \{1 - \lambda_i : \ i = 2, \ldots, p\}$.

Proof. Such a basis exists because $\mathbf{L}\mathbf{1} = \mathbf{0}$. By (2), $\mathbf{B}\mathbf{v}_1 = \mathbf{v}_1 - \mathbf{0} - \mathbf{v}_1 = \mathbf{0}$. For $i \geq 2$, $\mathbf{v}_i \perp \mathbf{1}$, so the last term of (2) vanishes and $\mathbf{B}\mathbf{v}_i = \mathbf{v}_i - \lambda_i \mathbf{v}_i$. □

In words: apart from the direction $\mathbf{1}$ ("all stocks move together"), which both matrices ignore, $\mathbf{B}$ is $\mathbf{L}$ reflected through $\frac{1}{2}$. Cheap cuts of the Laplacian become valuable splits for modularity.

Proposition 1. *Under* $\mathbf{d} = \mathbf{1}$*:*

(a) $\operatorname{rank}(\mathbf{L}) = p - k$ *if and only if* $\mathbf{B}$ *has eigenvalue 1 with multiplicity exactly* $k - 1$*. Moreover* $\mu_1 \leq 1$*.*

(b) For every $k \geq 2$*,*

$$\underbrace{\min_{\mathbf{V}^\top\mathbf{V} = \mathbf{I}_k} \operatorname{tr}(\mathbf{V}^\top \mathbf{L} \mathbf{V})}_{\textit{Fan term}} = (k-1) - \underbrace{\max_{\mathbf{U}^\top\mathbf{U} = \mathbf{I}_{k-1}} \operatorname{tr}(\mathbf{U}^\top \mathbf{B} \mathbf{U})}_{\textit{relaxed modularity}} + \max\{0, \lambda_k - 1\}.$$

If $\lambda_k \leq 1$ *the correction vanishes, which is the proposal's statement.*

(c) $\mu_1 = \max\{0, 1 - \lambda_2\}$*.*

Proof. (a) $\operatorname{rank}(\mathbf{L}) = p - k$ means exactly k zero eigenvalues, the first being λ_1, so exactly $k - 1$ of $\lambda_2, \ldots, \lambda_p$ are 0. By Lemma 1 these, and only these, give eigenvalue $1 - 0 = 1$ of $\mathbf{B}$; the remaining eigenvalue, on $\mathbf{v}_1$, is $0 \neq 1$. Since every $\lambda_i \geq 0$, every $1 - \lambda_i \leq 1$.

(b) By Ky Fan's theorem the left side is $\sum_{i \leq k} \lambda_i$ and the maximum on the right is the sum of the $k - 1$ largest elements of $\{0\} \cup \{x_i\}_{i \geq 2}$, with $x_i = 1 - \lambda_i$ decreasing. If $x_k \geq 0$ these are $x_2, \ldots, x_k$. If $x_k < 0$, the 0 is larger and displaces x_k. In both cases the sum is $\sum_{i=2}^{k} x_i + \max\{0, -x_k\} = (k-1) - \sum_{i \leq k} \lambda_i + \max\{0, \lambda_k - 1\}$, using $\lambda_1 = 0$. Rearranging gives the identity.

(c) The largest element of $\{0\} \cup \{1 - \lambda_i\}_{i \geq 2}$ is $\max\{0, 1 - \lambda_2\}$. □

When Algorithm 1's component requirement is met exactly, $\lambda_1 = \cdots = \lambda_k = 0$, so $\lambda_k \leq 1$ holds automatically and (b) holds with no correction. The condition matters for the intermediate iterates and for approximate solutions.

3.3 Why (b) needs $\lambda_k \leq 1$

For a unit vector $\mathbf{v} \perp \mathbf{1}$, (2) gives $\mathbf{v}^\top \mathbf{B} \mathbf{v} = 1 - \mathbf{v}^\top \mathbf{L} \mathbf{v}$. So $1 - \lambda_i$ is the relaxed modularity of splitting the graph along eigenvector i, while not splitting scores 0. If $\lambda_i > 1$, that split is worse than not splitting at all. The condition $\lambda_k \leq 1$ therefore says there are at least $k - 1$ worthwhile splits. When it fails, the Fan term still asks for k groups while modularity prefers fewer, and the two sides of (b) differ by exactly $\lambda_k - 1$. Figure 6 shows two unit-degree graphs where this happens.

Remark. Claim (c) inherits the same cut-off: $\max\{0, 1 - \lambda_2\}$ is 0 for every graph with $\lambda_2 \geq 1$, so on such graphs it carries no information. This matters for the time-varying graphs in Section 3.7.

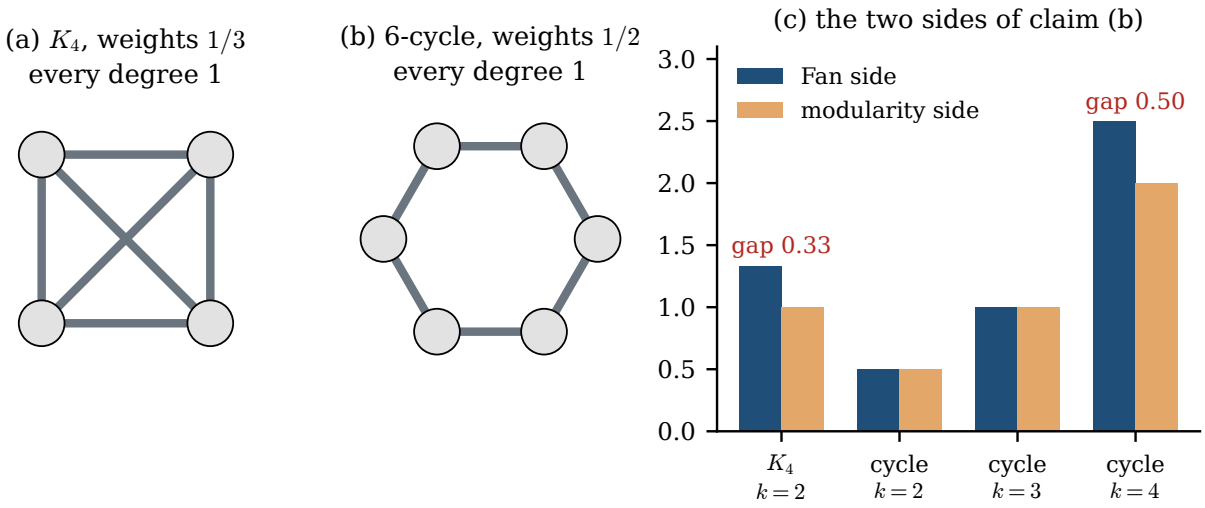


Figure 6: Two unit-degree counterexamples to (b) without its condition. (a) The complete graph on 4 nodes with weights 1/3 has $\lambda_2 = 4/3 > 1$: it fails already at $k = 2$. (b) The 6-cycle with weights 1/2 has $\mathrm{spec}(\mathbf{L}) = \{0, 0.5, 0.5, 1.5, 1.5, 2\}$: it satisfies (b) for $k = 2, 3$ and fails at $k = 4$, where $\lambda_4 = 1.5$. (c) The Fan side $\sum_{i \le k} \lambda_i$ against the modularity side $(k-1) - \sum_{i<k} \mu_i$; where they differ, the gap is exactly $\lambda_k - 1$.

3.4 When do the Laplacian and modularity agree in general?

For general degrees, $\mathbf{B} = \mathbf{D} - \mathbf{L} - \mathbf{d}\mathbf{d}^\top/2m$. Each matrix has a direction it treats as uninformative: $\mathbf{L}$ ignores $\mathbf{1}$ (everything moving together), while the null term of $\mathbf{B}$ points along $\mathbf{d}$ (each node moving in proportion to its degree). The two share eigenvectors when the diagonal part is a multiple of the identity and the two uninformative directions agree (Table 1).

setting	ignored by $\mathbf{L}$	ignored by $\mathbf{B}$	same eigenvectors?
regular, all degrees c (includes unit degree)	$\mathbf{1}$	$\mathbf{1}$	yes: $\mathbf{B} = c\mathbf{I} - \mathbf{L} - \frac{c}{p}\mathbf{1}\mathbf{1}^\top$
normalised pair, any degrees	$\sqrt{\mathbf{d}}$	$\sqrt{\mathbf{d}}$	yes: $\mathbf{B}_n = \mathbf{I} - \mathbf{L}_{\mathrm{sym}} - \mathbf{u}\mathbf{u}^\top$
plain pair, unequal degrees	$\mathbf{1}$	$\mathbf{d}$	no

Table 1: Three settings; $\mathbf{u} = \sqrt{\mathbf{d}/2m}$. The normalised identity is the 8 September result, and Part I is its special case $\mathbf{d} = \mathbf{1}$.

Regular. If $\mathbf{d} = c\mathbf{1}$ then $\mathbf{d}\mathbf{d}^\top/2m = \frac{c}{p}\mathbf{1}\mathbf{1}^\top$ and the proof of Lemma 1 goes through with 1 replaced by c: $\mathrm{spec}(\mathbf{B}) = \{0\} \cup \{c - \lambda_i\}$.

Normalised. For any positive degrees, since $\mathbf{D}^{-1/2}\mathbf{d} = \sqrt{\mathbf{d}}$, $\mathbf{B}_n = \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2} - \mathbf{u}\mathbf{u}^\top = \mathbf{I} - \mathbf{L}_{\mathrm{sym}} - \mathbf{u}\mathbf{u}^\top$, and $\mathbf{L}_{\mathrm{sym}}\mathbf{u} = \mathbf{0}$. Lemma 1 applies verbatim with $\mathbf{u}$ in place of $\mathbf{1}/\sqrt{p}$. So claim (a) holds for the normalised pair *whatever the degrees*.

Unequal degrees. For an eigenvector $\mathbf{v}$ of $\mathbf{L}$, $\mathbf{B}\mathbf{v} = \mathbf{D}\mathbf{v} - \lambda\mathbf{v} - \mathbf{d}(\mathbf{d}^\top\mathbf{v})/2m$, which is not a multiple of $\mathbf{v}$ in general. Figure 7 shows the smallest example I found.

Normalisation is not a free fix

Normalising changes coordinates, not the graph, but it changes the statistical model: Paper 1 uses $\mathbf{L}$ itself as the precision matrix, and using $\mathbf{L}_{\mathrm{sym}}$ there would be a different model. Inside Paper 1's framework, the only place modularity could add information is the plain pair with unequal degrees, that is, after dropping the unit-degree rule (Part III).

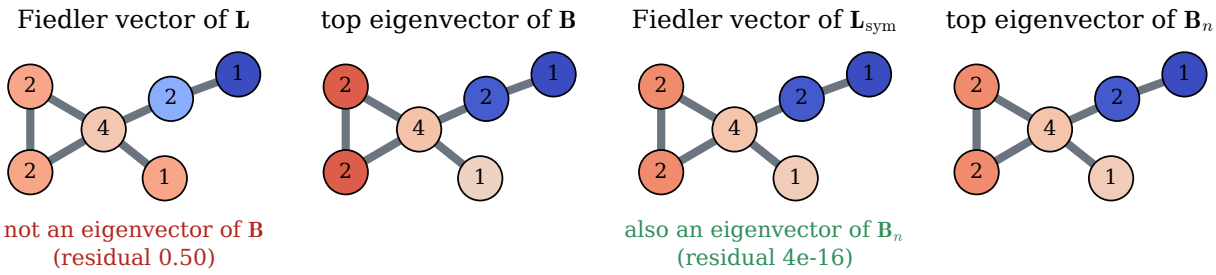


Figure 7: A triangle with two tails; numbers are degrees, colours are eigenvector entries. The Fiedler vector of $\mathbf{L}$ is not an eigenvector of $\mathbf{B}$ (residual $\|\mathbf{Bv} - (\mathbf{v}^\top\mathbf{Bv})\mathbf{v}\| = 0.50$), while the normalised pair matches to 4.0×10^{-16}. On a graph this small the sign pattern, and so the split, is still the same: different eigenvectors do not automatically mean different clusters.

3.5 The check on Paper 1's learned graph

The test case is Algorithm 1 on the S&P data, the graph behind Paper 1's Fig. 2 (Figure 8). Its three components have 52, 28 and 17 stocks and are exactly the three sectors (ARI 1.000). Because the degree rule is a penalty, the learned degrees lie in [1.0012, 1.0069] rather than at exactly 1, so every prediction of Proposition 1 should hold approximately, with an error on the scale of that tolerance. Table 2 and Figure 9 compare them.

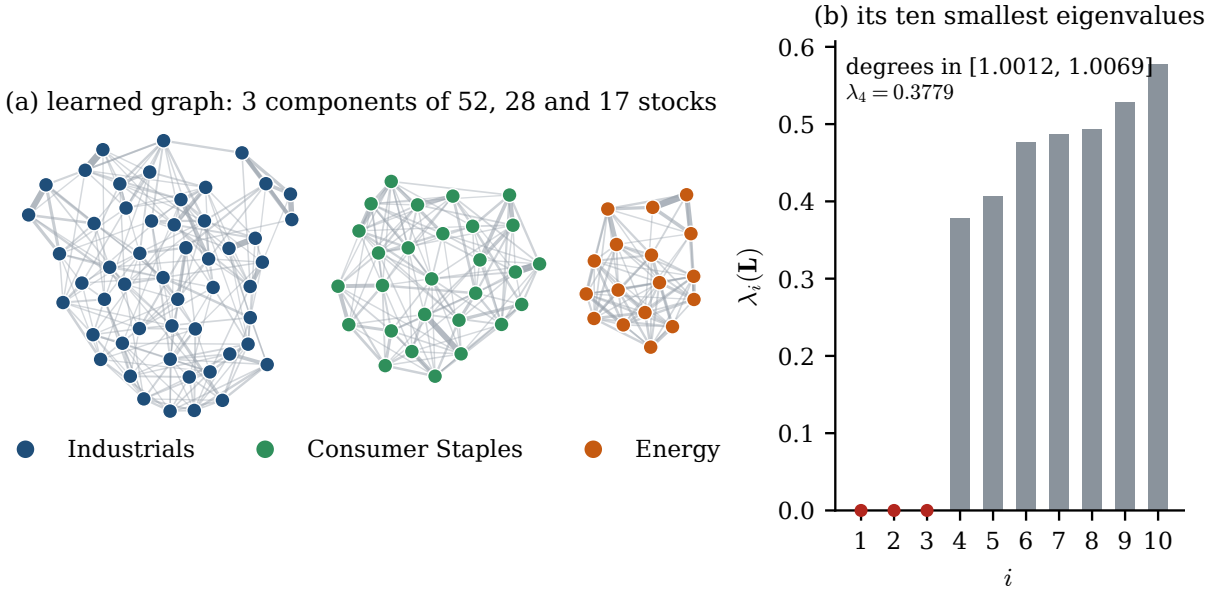


Figure 8: (a) The graph learned by Algorithm 1 from the S&P data, coloured by sector; edge width is weight, and the strongest 450 of its 1,194 edges are drawn. Each component is exactly one sector. (b) Its Laplacian has three zero eigenvalues (one per component) and then a gap.

Verdict: reproduced, within the solver's tolerance. The partition prediction holds exactly, and the spectral predictions hold to within the degree deviation. Most of that deviation is a uniform shift: every degree is slightly above 1, so the graph is closer to c-regular with $c = 1.0047$ than to unit degree. This also explains why μ_1, μ_2 slightly exceed the ceiling of 1 in (a), and why (b) misses by $-0.0093 \approx 2(1 - \bar{d})$.

3.6 Loosening the degree rule

To see the identity fail as degrees spread, I reran Algorithm 1 on the same data with the same settings, lowering only the degree penalty ρ. Every run still returns three components and no

quantity	Part I predicts	learned graph
degrees d_i	all 1	in [1.0012, 1.0069], mean 1.0047
$\|\mathbf{B} - (\mathbf{I} - \mathbf{L} - \mathbf{1}\mathbf{1}^\top/p)\|_F$	0	0.0479 (1.7% of $\|\mathbf{B}\|_F$)
three smallest $\lambda_i(\mathbf{L})$	0, 0, 0	all below 1.3×10^{-15}; next 0.3779
largest $\mu_i(\mathbf{B})$, claim (a)	1, 1, then $1 - \lambda_4$	1.0052, 1.0041, then 0.6274
claim (b): $\sum_{i \le 3} \lambda_i$ against $2 - \mu_1 - \mu_2$	equal	6.7×10^{-16} against -0.0093
largest $\lvert\lambda_i - (1 - \mu_i)\rvert$ over the 96 pairs	0	0.0059 (mean 0.0047)
same, reflecting about the mean degree	0	0.0020 (mean 0.0006)
k-means on the top 2 eigenvectors of $\mathbf{B}$	the components	ARI 1.000 against components and against sectors

Table 2: Part I's predictions on the learned graph. "Reflecting about the mean degree" uses the regular form of Table 1 with $c = \bar{d} = 1.0047$.

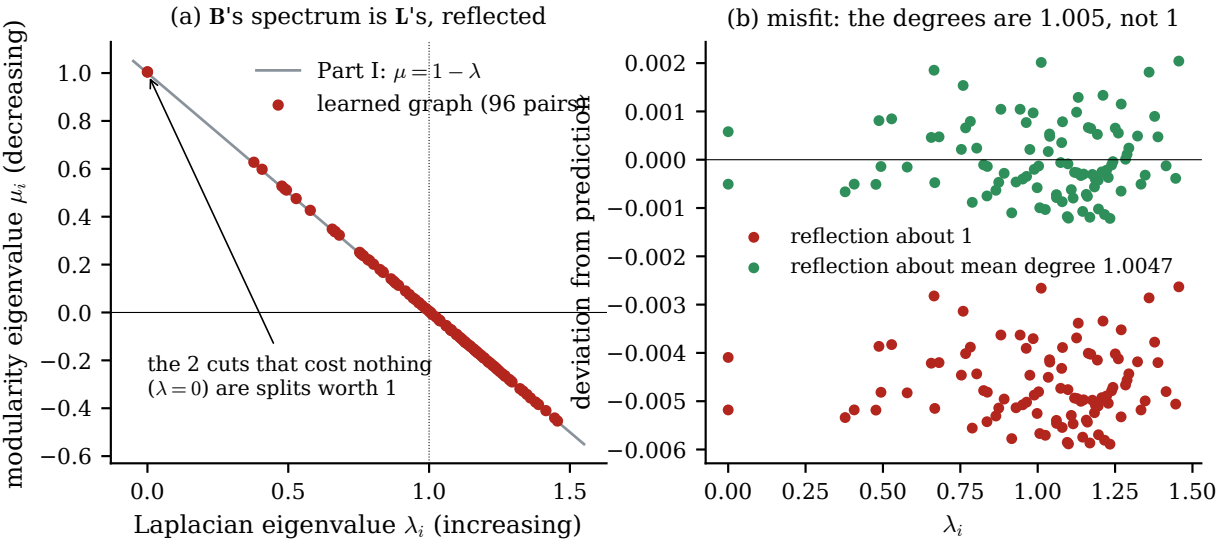


Figure 9: (a) Each eigenvalue pair of the learned graph (on the directions orthogonal to $\mathbf{1}$) lies on the line $\mu = 1 - \lambda$, as Lemma 1 predicts. (b) The remaining misfit, magnified. Reflecting about 1 leaves a uniform offset of about -0.005 (red); reflecting about the mean degree 1.0047 removes it (green), leaving only the spread of the degrees around their mean.

isolated stock (Figure 10).

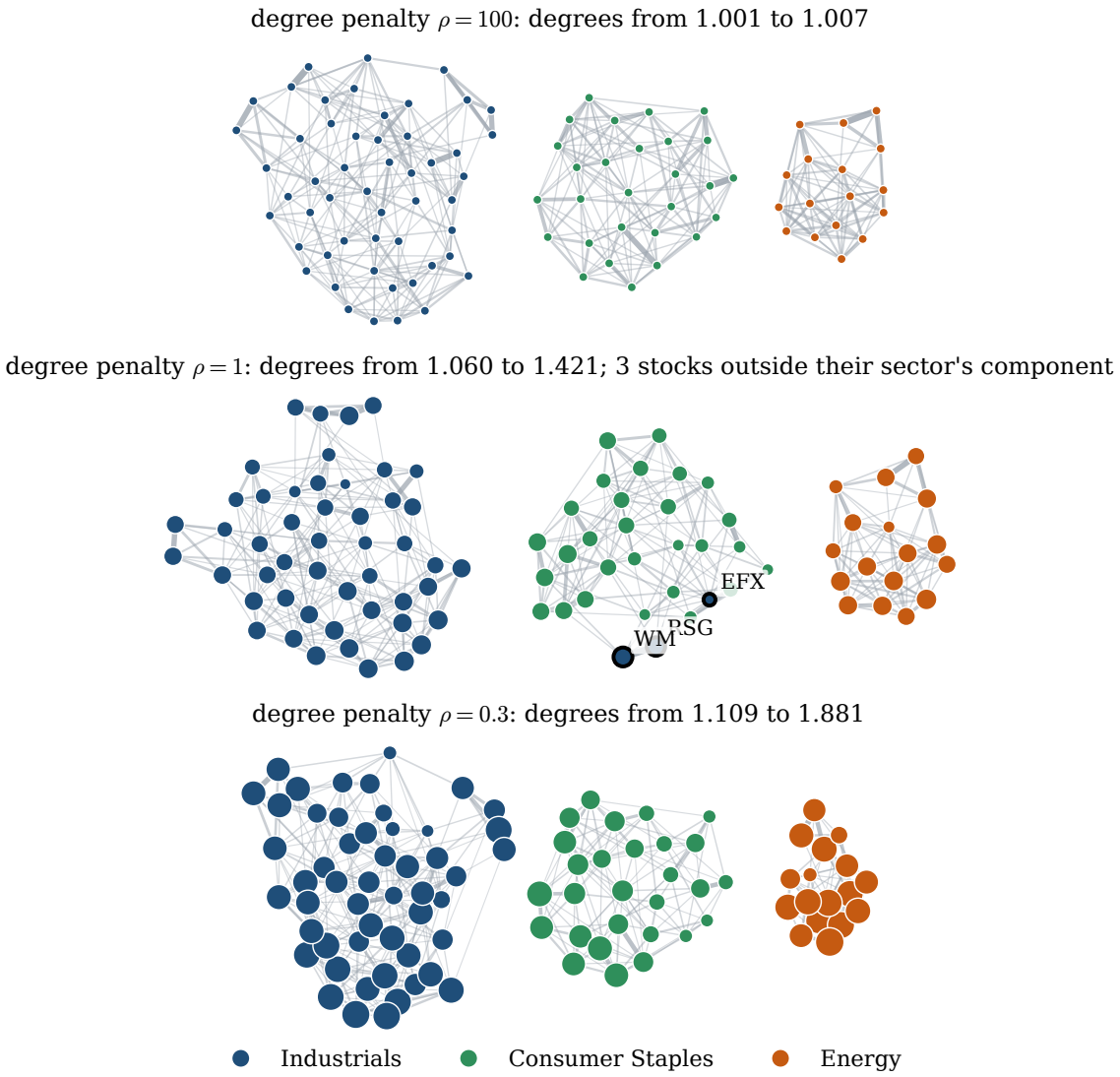


Figure 10: Algorithm 1 at three strengths of the degree rule; node size grows with degree. At $\rho = 100$ every degree is essentially 1. At $\rho = 0.3$ they range up to 1.881, yet the components are still the sectors. At $\rho = 1$ three Industrials stocks (outlined: EFX, RSG and WM) join the Consumer Staples component.

Three findings:

- **The algebra of Part I degrades steadily** as degrees spread (degree CV from 0.0014 to 0.126): the residual grows from 1.7% to more than the size of $\mathbf{B}$, and the top eigenvalues of $\mathbf{B}$ pass the ceiling of 1 in (a).
- **The normalised identity does not degrade at all**: $\mathbf{B}_n$ keeps eigenvalue 1 exactly twice at every ρ, as Table 1 predicts.
- **The partition survives**: even at $\rho = 0.3$, k-means on the plain $\mathbf{B}$ returns the components exactly. With three components and no edges between them, the eigenvector drift ($\sin \leq 0.24$) is too small to move any stock. I have no proof that this always happens.

The $\rho = 1$ run is out of line: EFX, RSG and WM join Consumer Staples (ARI against sectors 0.896). RSG and WM are waste-management companies and EFX a credit bureau, all commonly treated as defensive, so the grouping is plausible; the runs on either side return the sectors exactly. This is a property of that learned graph; the Part I comparison still matches its components

ρ	degree range	degree CV	components	rel. residual	μ_1, μ_2 of $\mathbf{B}$	sin	ARI vs comp.	μ_1, μ_2 of $\mathbf{B}_n$
100	[1.001, 1.007]	0.0014	52/28/17	0.017	1.005, 1.004	0.002	1.000	1, 1
30	[1.004, 1.023]	0.0047	52/28/17	0.056	1.017, 1.013	0.008	1.000	1, 1
10	[1.011, 1.064]	0.0129	52/28/17	0.152	1.048, 1.038	0.021	1.000	1, 1
3	[1.031, 1.184]	0.0344	52/28/17	0.392	1.139, 1.110	0.058	1.000	1, 1
1	[1.060, 1.421]	0.0700	49/31/17	0.730	1.318, 1.254	0.123	1.000	1, 1
0.3	[1.109, 1.881]	0.1265	52/28/17	1.098	1.647, 1.491	0.236	1.000	1, 1

Table 3: The sweep. Degree CV is the coefficient of variation (standard deviation over mean) of the degrees. Relative residual is $\|\mathbf{B} - (\mathbf{I} - \mathbf{L} - \mathbf{1}\mathbf{1}^\top/p)\|_F / \|\mathbf{B}\|_F$. sin is the largest principal angle between the zero-eigenspace of $\mathbf{L}$ and the span of $\mathbf{1}$ and the top two eigenvectors of $\mathbf{B}$ (0 means the same subspace). ARI compares k-means on the top two eigenvectors of $\mathbf{B}$ with the components. The eigenvalues of $\mathbf{B}_n$ equal 1 to six decimals in every row.

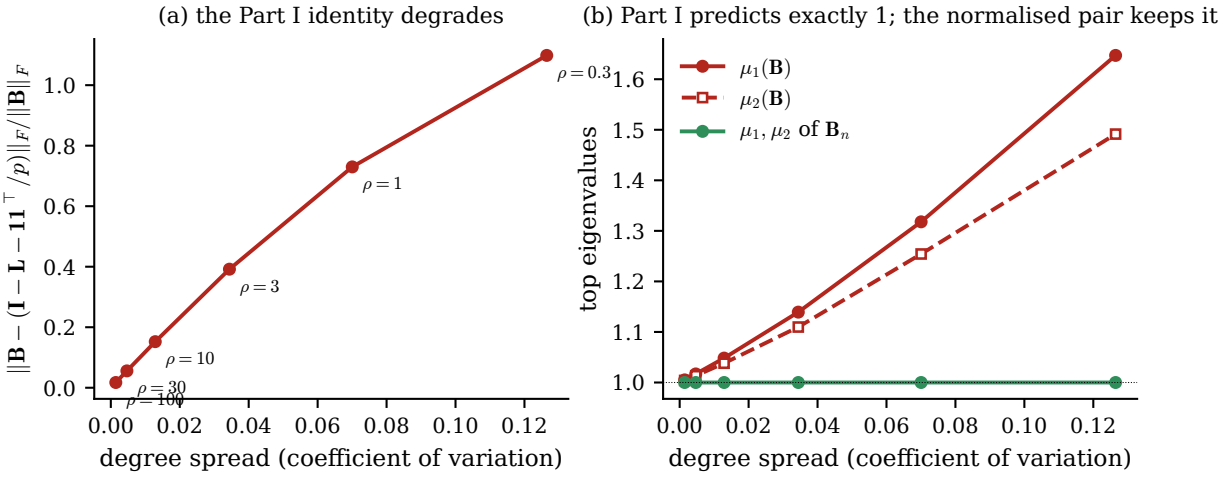


Figure 11: The sweep as degree spread grows. (a) The Part I identity degrades steadily, reaching a residual larger than $\mathbf{B}$ itself. (b) Part I predicts the top two eigenvalues of $\mathbf{B}$ are exactly 1; they climb to 1.65. The normalised pair keeps them at exactly 1 throughout.

exactly.

3.7 Where claim (c) applies

Claim (c) is exact under unit degree, but Paper 1's stress indicator comes from its time-varying learner (its Problem (11)), which has no degree rule. Our 243 rolling-window graphs for the FAAMUNG stocks (18 October note) have mean degree 2.13 and degree CV 0.305, more than twice the largest spread in Table 3, and $\lambda_2 \geq 1$ in 58% of windows (140 of 243), where $\max\{0, 1 - \lambda_2\}$ is 0. On those graphs μ_1 and λ_2 are different quantities, so (c) does not connect the trading indicator to modularity. If a modularity indicator on the time-varying graphs is still of interest, the normalised identity $\mu_1(\mathbf{B}_n) = \max\{0, 1 - \lambda_2(\mathbf{L}_{\text{sym}})\}$ holds for any degrees and could be the version to test.

3.8 Part I conclusions

Under the unit-degree rule the modularity matrix is a reflection of the Laplacian, so it can neither find different clusters nor count them differently, and Algorithm 1's Fan term is already a relaxed modularity. Paper 1's learned graph confirms this to within the solver's tolerance. Modularity could only add information once degrees are allowed to differ, which is Part III's direction; and the choice of null model, which is Part II.

4 Part II: the market as a built-in null model

4.1 The question

On most days the whole market rises or falls together. That shared movement explains a large part of every stock's return, and it says nothing about which stocks belong together; it is why nearly every entry of $\mathbf{S}$ is positive (Figure 2). Modularity-based methods for correlation matrices therefore subtract an explicit model of it before looking for groups. Paper 1 never does, yet it recovers the sectors. Part II asks why, and whether it is always safe.

One-factor model, beta and loading
The *one-factor (market) model* writes each return as $x_{t,i} = b_i f_t + (\text{sector part}) + (\text{noise})$, where f_t is the market's return on day t. The *beta* b_i says how strongly stock i follows the market: 1 is average, above 1 amplifies the market, below 1 dampens it. In a correlation matrix the relevant quantity is the *market loading* $b_i = \text{corr}(x_i, f)$; I use both, and call both "beta" where the difference does not matter. Here the market is the S&P 500 index (`^GSPC`).

4.2 The learner removes the market on its own

Let $\mathbf{J} = \mathbf{I} - \mathbf{1}\mathbf{1}^\top/p$. Applying $\mathbf{J}$ to a vector subtracts its average, so $\mathbf{JSJ}$ is the correlation structure of the returns after *each day's average across stocks* has been removed.

Proposition 2. *For every Laplacian* $\mathbf{L}$ *and every symmetric* $\mathbf{S}$, $\text{tr}(\mathbf{LS}) = \text{tr}(\mathbf{LJSJ})$. *Hence Problem (1), Algorithm 1 and any learner whose only data term is* $\text{tr}(\mathbf{LS})$ *learn the same graph from* $\mathbf{S}$ *and from* $\mathbf{JSJ}$.

Proof. $\mathbf{L1} = \mathbf{0}$ gives $\mathbf{LJ} = \mathbf{L}$ and, by symmetry, $\mathbf{JL} = \mathbf{L}$. So $\text{tr}(\mathbf{LJSJ}) = \text{tr}(\mathbf{JLJS}) = \text{tr}(\mathbf{LS})$. The other terms of the objectives depend on $\mathbf{L}$ alone. □

So the learner works with the de-averaged data whether we ask it to or not. What does that remove? Take the one-factor model, $\mathbf{S} = \sigma^2 \mathbf{b}\mathbf{b}^\top + \mathbf{S}_0$, with $\mathbf{S}_0$ the sector and noise parts. By (1),

the market's share of the data fit is

$$\operatorname{tr}(\mathbf{L}\,\sigma^2\mathbf{b}\mathbf{b}^\top) = \sigma^2\,\mathbf{b}^\top\mathbf{L}\mathbf{b} = \sigma^2\sum_{i<j} W_{ij}(b_i - b_j)^2. \tag{3}$$

- **Equal betas are removed exactly.** If every b_i is the same, (3) is 0 for every graph: the learner ignores the market completely, with no model of it.
- **Unequal betas leave a penalty.** Otherwise each link pays in proportion to the squared beta difference of its two stocks, so the learner is pushed to link stocks with similar betas. Equivalently, it sees $\sigma^2(\mathbf{J}\mathbf{b})(\mathbf{J}\mathbf{b})^\top$, which acts like an extra factor sorting stocks by beta.

Beta inequality $\sin^2\theta$

The total size of the penalty over all pairs is

$$\sum_{i<j}(b_i - b_j)^2 = p\sum_i b_i^2 - \Big(\sum_i b_i\Big)^2 = p\|\mathbf{J}\mathbf{b}\|^2 = p\|\mathbf{b}\|^2\sin^2\theta, \qquad \sin^2\theta = 1 - \frac{(\mathbf{1}^\top\mathbf{b})^2}{p\|\mathbf{b}\|^2},$$

where θ is the angle between the beta vector and $\mathbf{1}$. $\sin^2\theta$ is 0 when all betas are equal and grows as they spread; it does not depend on the overall size of the betas. It is the measure of beta inequality used throughout Part II.

Whether the penalty helps or hurts depends on where the betas sit relative to the groups. If stocks in the same sector have similar betas, the penalty falls on links between sectors and *reinforces* the sectors. If betas vary inside sectors, it *competes* with them.

Explicit nulls compared

Four standard ways to remove the market before learning, plus one available only in simulation:

(i) *S-mode*: subtract the top eigenvector component $\mu_1\mathbf{u}_1\mathbf{u}_1^\top$ of $\mathbf{S}$ (the "market mode"), then rescale to unit diagonal;

(ii) *resid-SPX*: regress each stock on the S&P 500 index and use the correlation of the residuals;

(iii) *resid-EW*: the same, on the equal-weighted average of the 97 stocks;

(iv) *RMT*: the "group component" of MacMahon and Garlaschelli (2015). Random matrix theory says that for pure noise the eigenvalues of a correlation matrix stay below the *Marchenko-Pastur edge* $(1+\sqrt{p/T})^2$; the group component keeps only the eigenvalue modes above that edge, minus the market mode;

(v) *oracle* (simulation only): residuals on the true market factor.

4.3 Real data: the S&P panel

The loadings of the 97 stocks on the S&P 500 lie in [0.177, 0.743] (mean 0.513), so $\sin^2\theta = 0.064$; the regression betas lie in [0.347, 1.839]. The betas are largely a property of the sector: mean loadings are 0.363 for Consumer Staples, 0.597 for Industrials and 0.506 for Energy (mean regression betas 0.568, 1.042 and 1.192), and 57.7% of the variation in loadings is between sectors rather than within them (Figure 12). For the RMT null the Marchenko-Pastur edge is 1.847, and 4 group modes lie above it.

Table 4 runs both learners on all six inputs. Besides the ARI, it reports a *beta ratio*: the link-weighted average of $|b_i - b_j|$ over same-sector pairs, divided by the plain average over the same pairs. Values below 1 mean that within a sector the graph prefers to link stocks with similar betas, which is the penalty (3) at work.

- **The invariance holds numerically.** Graphs from $\mathbf{S}$ and from $\mathbf{JSJ}$ agree to 2.7×10^{-10} (Problem (1)) and 1.3×10^{-4} (Algorithm 1, within its stopping tolerance), and every metric is identical.

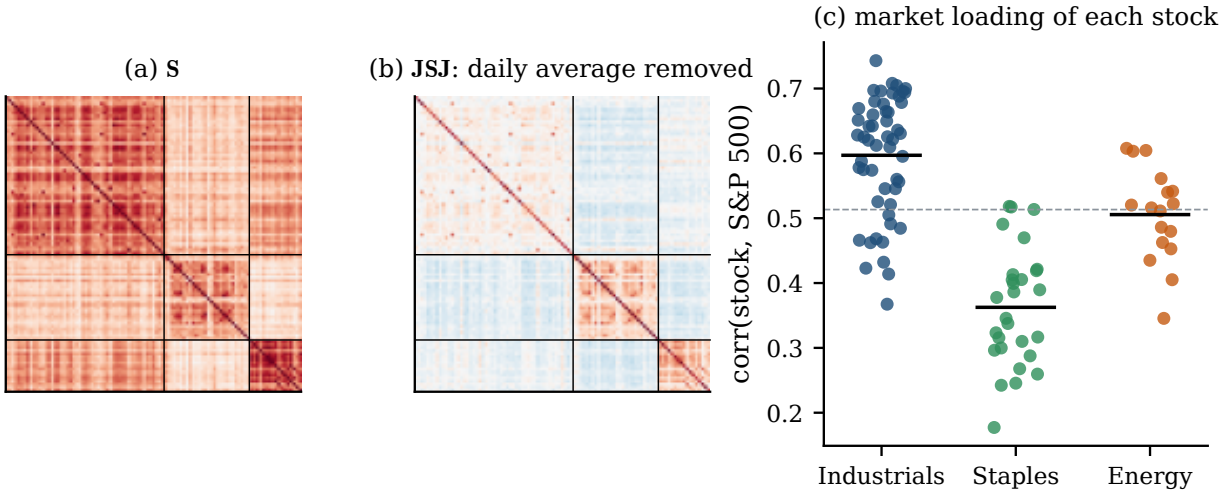


Figure 12: (a) The correlation matrix, sorted by sector. (b) After removing each day's average (**JSJ**), the uniform positive background is gone and the sector blocks stand out; this is what the learner actually sees. (c) Each stock's loading on the S&P 500, by sector (bars: sector means; dashed: overall mean). Consumer Staples are defensive, low-beta stocks.

	Algorithm 1 ($k = 3$)				Problem (1) + k-means		
input	comp.	ARI sectors	ARI beta	beta ratio	ARI sectors	ARI beta	beta ratio
S (built-in null)	3	**1.000**	0.277	0.786	**0.900**	0.295	0.862
JSJ	3	1.000	0.277	0.786	0.900	0.295	0.862
S-mode	1	0.830	0.257	0.820	0.867	0.292	0.879
resid-SPX	1	0.830	0.257	0.816	0.802	0.252	0.872
resid-EW	3	0.896	0.298	0.794	0.867	0.292	0.867
RMT	1	0.875	0.291	0.928	0.900	0.268	0.944

Table 4: The S&P panel. "comp." is the number of components Algorithm 1 returns; where it is 1, clusters come from k-means on the bottom three eigenvectors. ARI beta compares the clusters with a three-way split of the stocks by loading.

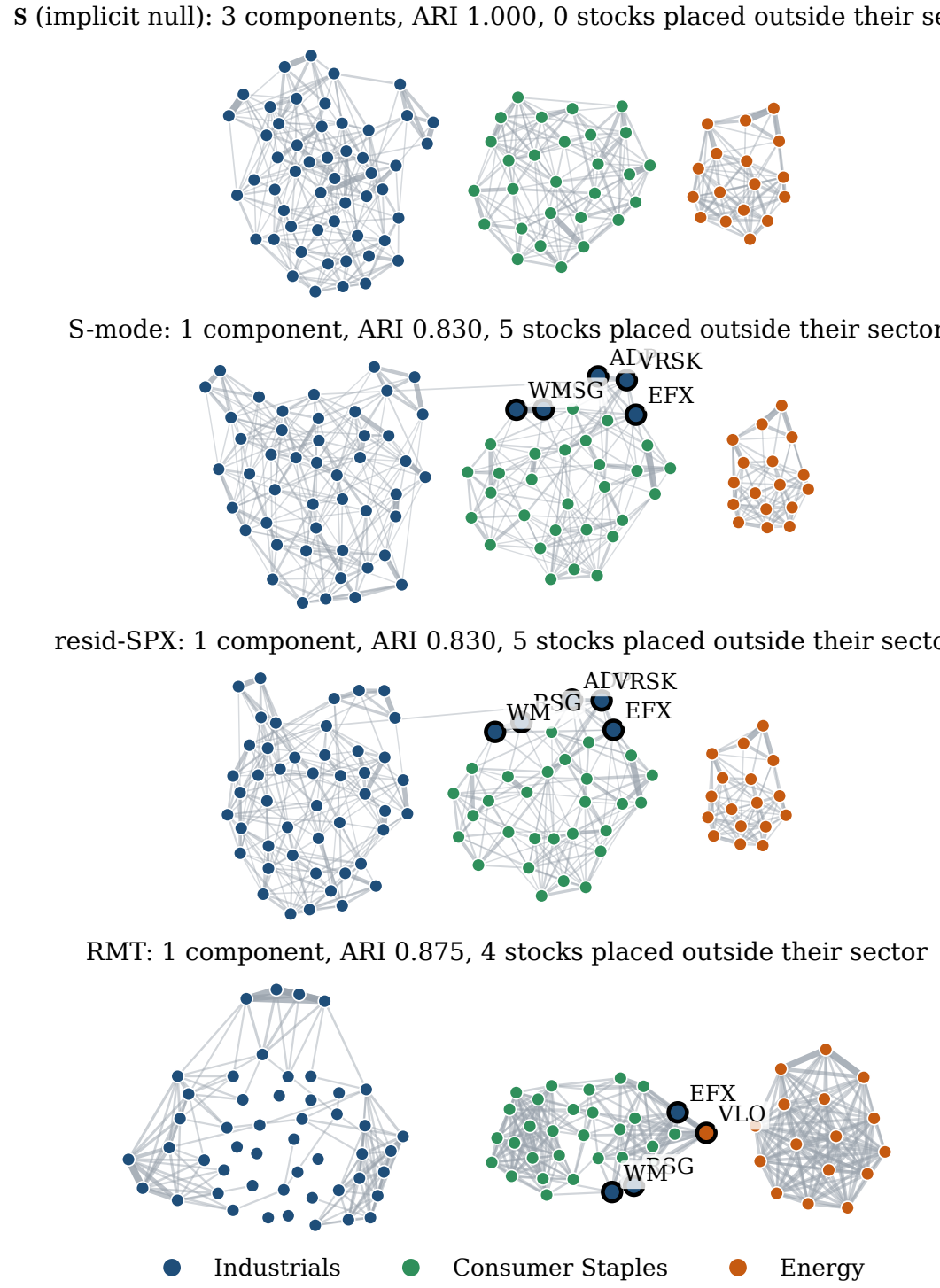


Figure 13: Algorithm 1's graphs from four of the inputs, coloured by sector, drawn with each detected cluster laid out separately. Outlined and labelled: stocks placed in a cluster dominated by another sector. The built-in null gets every stock right; the explicit nulls move the same handful of Industrials stocks into Consumer Staples (and, for RMT, the refiner VLO).

- **No explicit null beats the built-in one.** Under Algorithm 1 every explicit null loses: ARI 0.830 to 0.896 against 1.000. Under Problem (1) the RMT null ties at 0.900 and the rest lose (0.802 to 0.867). *Explicit market removal improving on the built-in null: not supported on this data.*
- **The same stocks move every time.** Apart from the refiner VLO, every misplaced stock is one of ADP, EFX, RSG, VRSK and WM (payroll, credit-data, analytics and waste-management companies: Industrials by classification, defensive by behaviour), and each joins Consumer Staples (Figure 13). VLO is misplaced by Problem (1) on every input, the built-in null included, and by Algorithm 1 on RMT.
- **A likely reason: betas carry sector information here.** With 57.7% of the loading variance between sectors, the penalty (3) falls mostly on links between sectors and supports the sector split. Removing the market removes that support, and the low-beta Industrials stocks drift towards the low-beta sector.
- **The penalty is visible inside sectors.** With the built-in null the beta ratio is 0.786 (Algorithm 1) and 0.862 (Problem (1)); every explicit null raises it, RMT the most (0.928, 0.944), as (3) predicts once the market term is gone. In absolute terms the market term is small: $\mathbf{b}^\top \mathbf{L}\mathbf{b}$ is 1.1% of $\mathrm{tr}(\mathbf{LS})$ on the Algorithm 1 graph and 1.7% on the Problem (1) graph.

4.4 Simulated markets: when betas cut across sectors

The S&P data cannot show what happens when betas vary *within* sectors, so I simulated it.

Model. $x_{t,i} = b_i f_t + s_{g(i),t} + e_{t,i}$, with a market factor $f_t \sim N(0,1)$, one factor per sector $s_{g,t} \sim N(0, 0.6^2)$ and noise $e_{t,i} \sim N(0, 1.2^2)$, all independent; $p = 60$ stocks in three sectors of 20, and $T = 753$ days as in the real data. Three beta designs (Figure 14): *equal* (every $b_i = 1$); *aligned* ($b_i = 1-\delta,\ 1,\ 1+\delta$ by sector); and *cross* (in every sector, half the stocks at $1+\delta$ and half at $1-\delta$). The population loadings are $b_i / \sqrt{b_i^2 + 0.6^2 + 1.2^2}$, and $\sin^2\theta$ is computed from them. Each simulated data set is learned from $\mathbf{S}$, from S-mode and from the oracle, with both learners.

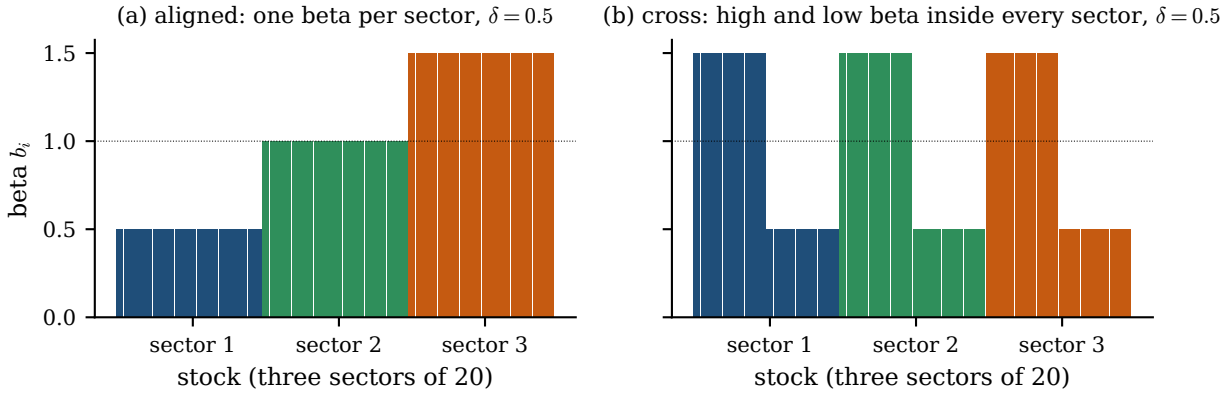


Figure 14: The two unequal-beta designs at $\delta = 0.5$. (a) Aligned: betas follow the sectors, as on the real data. (b) Cross: the beta split is orthogonal to the sectors, so the penalty (3) competes with them.

Coarse grid ($\delta = 0.2, 0.4, 0.6, 0.8$, two seeds; 108 runs).

- *Equal and aligned*: every input and learner recovers the sectors exactly (ARI 1.000) in every run, up to $\sin^2\theta = 0.219$. Unequal betas are harmless when they follow the groups. This is the real-data situation.
- *Cross*: exact at $\delta = 0.2$ and 0.4 ($\sin^2\theta = 0.017,\ 0.072$). At $\delta = 0.6$ and 0.8 ($\sin^2\theta = 0.173,\ 0.322$) both learners on $\mathbf{S}$ cluster by beta in both seeds: ARI 0.134 against the sectors and 0.773 against the two beta groups (not 1, because there are three clusters for two groups).

S-mode and the oracle stay at 1.000 (Figure 15).

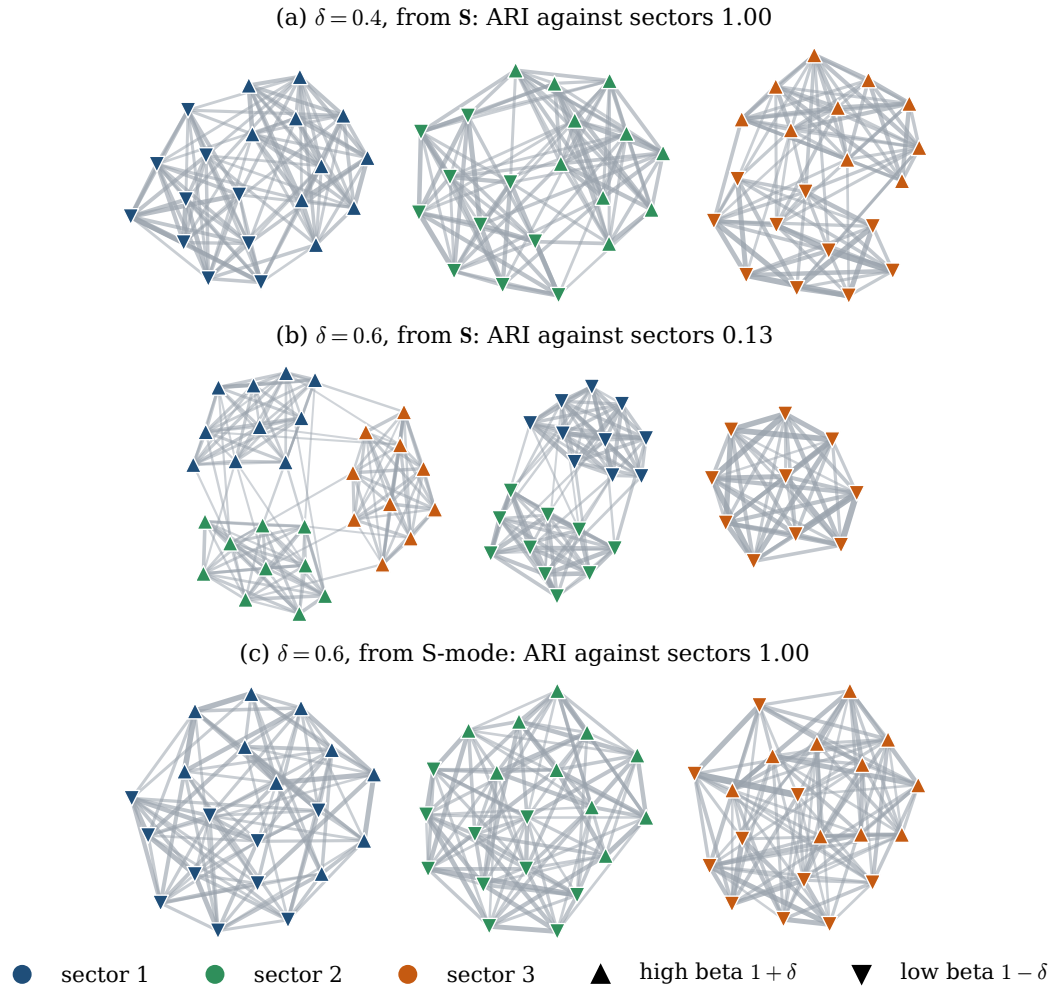


Figure 15: Algorithm 1 on simulated cross-design data (seed 0). Colour is the true sector, the triangle's direction the beta group. (a) At $\delta = 0.4$ the components are the sectors. (b) At $\delta = 0.6$ they are held together by beta instead: one component holds all 30 high-beta stocks from every sector, and the low-beta stocks split in two. (c) The same data with the market mode removed: the sectors again.

Fine grid (cross design, δ from 0.400 to 0.600 in steps of 0.025, five seeds; 270 runs). Table 5 and Figure 16 locate the switch. S-mode and the oracle give ARI 1.000 in all 180 of their runs.

The switch is sharp. For Problem (1) it begins between $\sin^2\theta = 0.104$ and 0.116 and is complete by 0.143; for Algorithm 1 it begins between 0.129 and 0.143 and is complete by 0.173. Algorithm 1 holds out longer; I have not tested why. Where the switch happens depends on how strong the sector factor is relative to the noise, which here is our choice (0.6 against 1.2). So these values belong to this design, and the real data's $\sin^2\theta = 0.064$ cannot be compared with them directly. Section 4.6 gives a comparison that does carry over.

4.5 A perturbation bound

The proposal asks for a bound that links beta inequality to how far the cluster eigenvectors can move. I derive it for Problem (1), which is convex; Algorithm 1 is not, and the argument does not apply to it. Two more terms are needed.

		Algorithm 1 on **S**		Problem (1) on **S**	
δ	$\sin^2\theta$	ARI sectors	seeds by beta	ARI sectors	seeds by beta
0.400	0.0715	1.000	0/5	1.000	0/5
0.425	0.0815	1.000	0/5	1.000	0/5
0.450	0.0921	1.000	0/5	1.000	0/5
0.475	0.1036	1.000	0/5	1.000	0/5
0.500	0.1158	1.000	0/5	0.767	1/5
0.525	0.1288	1.000	0/5	0.346	3/5
0.550	0.1426	0.767	1/5	0.134	5/5
0.575	0.1572	0.459	2/5	0.134	5/5
0.600	0.1726	0.134	5/5	0.134	5/5

Table 5: The switch from sectors to beta groups, cross design. ARI is the mean over five seeds; "seeds by beta" counts the seeds whose clusters match the beta groups better than the sectors.

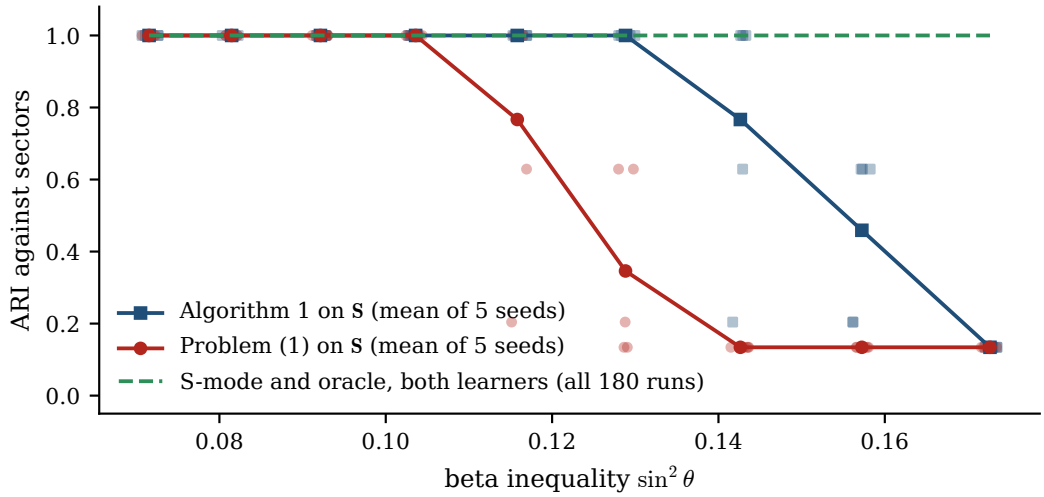


Figure 16: ARI against the sectors as beta inequality grows (cross design; faint markers are single seeds, lines are means). The built-in null switches to clustering by beta over a narrow range; the explicit nulls never do.

Principal angles and $\|\sin\Theta\|_F$

To compare two k-dimensional subspaces (here, the spans of the bottom k eigenvectors of two Laplacians), one uses their k *principal angles* $\theta_1, \dots, \theta_k$: the smallest angles that can be achieved between unit vectors, one from each subspace, chosen one pair at a time. $\|\sin\Theta\|_F = (\sum_i \sin^2\theta_i)^{1/2}$ is 0 for identical subspaces and at most $\sqrt{k}$.

Strong convexity and the Davis-Kahan theorem

A function is μ-*strongly convex* if its curvature is at least μ in every direction; then a small change in its linear part can move the minimiser only a little. The *Davis-Kahan theorem* is the classic statement that the eigenvectors of a symmetric matrix move little under a perturbation that is small compared with the eigengap separating them from the others.

In the edge weights $\mathbf{w} = (W_{ij})_{i<j} \geq 0$, Problem (1) reads $\min_{\mathbf{w}\geq 0}\langle \mathbf{w}, \mathbf{z}(\mathbf{S}) + \beta\mathbf{1}\rangle + g(\mathbf{w})$, with $z_{ij} = S_{ii} + S_{jj} - 2S_{ij}$ and $g(\mathbf{w}) = -\log\operatorname{gdet}\mathbf{L}(\mathbf{w})$. Adding the market term $\mathbf{b}\mathbf{b}^\top$ to $\mathbf{S}$ adds $\Delta z_{ij} = (b_i - b_j)^2$ to $\mathbf{z}$: a linear change, of total size $\|\Delta\mathbf{z}\|_1 = p\|\mathbf{b}\|^2\sin^2\theta$. Let $\mathbf{w}_c$ solve the clean problem (from $\mathbf{S} - \mathbf{b}\mathbf{b}^\top$) and $\mathbf{w}_o$ the observed one (from $\mathbf{S}$).

Proposition 3 (Step 1)**.** *Suppose both solutions are connected graphs and let M be the larger of their largest Laplacian eigenvalues. Then*

$$\langle \Delta\mathbf{z}, \mathbf{w}_o - \mathbf{w}_c\rangle \leq 0, \qquad \|\mathbf{w}_o - \mathbf{w}_c\| \leq \frac{M^2}{2}\|\Delta\mathbf{z}\|, \qquad \|\mathbf{L}(\mathbf{w}_o) - \mathbf{L}(\mathbf{w}_c)\|_F \leq \sqrt{2p}\,\|\mathbf{w}_o - \mathbf{w}_c\|.$$

Proof. Curvature. For a connected $\mathbf{L}$ and a direction $\mathbf{h}$, the second derivative of g is $\operatorname{tr}(\mathbf{L}^+\mathbf{L}(\mathbf{h})\mathbf{L}^+\mathbf{L}(\mathbf{h}))$, with $\mathbf{L}^+$ the pseudo-inverse. $\mathbf{L}(\mathbf{h})$ acts on the directions orthogonal to $\mathbf{1}$, where $\mathbf{L}^+ \succeq \lambda_{\max}^{-1}\mathbf{I}$, so the second derivative is at least $\|\mathbf{L}(\mathbf{h})\|_F^2/\lambda_{\max}^2$, and $\|\mathbf{L}(\mathbf{h})\|_F^2 = \sum_i(\sum_j h_{ij})^2 + 2\|\mathbf{h}\|^2 \geq 2\|\mathbf{h}\|^2$. Every graph on the segment between $\mathbf{w}_c$ and $\mathbf{w}_o$ is connected and, since $\lambda_{\max}$ is convex in $\mathbf{w}$, has $\lambda_{\max} \leq M$. So g is μ-strongly convex there with $\mu = 2/M^2$.

Comparison. Optimality over $\mathbf{w} \geq 0$ gives $\langle \mathbf{z} + \Delta\mathbf{z} + \beta\mathbf{1} + \nabla g(\mathbf{w}_o), \mathbf{w}_c - \mathbf{w}_o\rangle \geq 0$ and $\langle \mathbf{z} + \beta\mathbf{1} + \nabla g(\mathbf{w}_c), \mathbf{w}_o - \mathbf{w}_c\rangle \geq 0$. Adding them,

$$\mu\|\mathbf{w}_o - \mathbf{w}_c\|^2 \leq \langle \nabla g(\mathbf{w}_o) - \nabla g(\mathbf{w}_c), \mathbf{w}_o - \mathbf{w}_c\rangle \leq \langle \Delta\mathbf{z}, \mathbf{w}_c - \mathbf{w}_o\rangle \leq \|\Delta\mathbf{z}\|\,\|\mathbf{w}_o - \mathbf{w}_c\|.$$

The middle inequality gives the sign statement, and dividing gives the second claim. For the third, Cauchy-Schwarz gives $(\sum_j h_{ij})^2 \leq (p-1)\sum_j h_{ij}^2$; summing over i and adding $2\|\mathbf{h}\|^2$ gives at most $2p\|\mathbf{h}\|^2$. □

The sign statement reads plainly: compared with the clean graph, the observed graph has, in aggregate, moved weight away from pairs with different betas.

Step 2 is the Davis-Kahan theorem in the form of Yu, Wang and Samworth (2015, Theorem 2). With $\text{gap} = \lambda_{k+1}(\mathbf{L}_c) - \lambda_k(\mathbf{L}_c) > 0$, the bottom-$k$ eigenvector subspaces satisfy $\|\sin\Theta\|_F \leq 2\sqrt{k}\,\|\Delta\mathbf{L}\|_2/\text{gap}$, where $\|\cdot\|_2$ is the largest singular value. Chaining both steps through $\|\Delta\mathbf{L}\|_2 \leq \|\Delta\mathbf{L}\|_F$,

$$\|\sin\Theta\|_F \leq \frac{\sqrt{2kp}\,M^2\,\|\Delta\mathbf{z}\|}{\text{gap}}, \qquad \|\Delta\mathbf{z}\| \leq p\|\mathbf{b}\|^2\sin^2\theta. \tag{4}$$

This is the shape the proposal anticipated: the cluster subspace moves by at most a constant times $\sin^2\theta/\text{gap}$. Since $\|\sin\Theta\|_F \leq \sqrt{k}$ always, (4) only says something when its right side is below $\sqrt{k}$.

4.6 The bound against the measurements

Table 6 evaluates each step for Problem (1) on the cross design (mean over three seeds) and on the real data. The clean graph is learned from $\mathbf{S} - \mathbf{b}\mathbf{b}^\top$, using the population loadings (simulation) or the sample loadings on the S&P 500 (real). M is between 0.189 and 0.195 in every row.

		Step 1: $\|\Delta\mathbf{w}\|$			$\|\Delta\mathbf{L}\|$					Step 2: DK		
δ	$\sin^2\theta$	actual	bound	ratio	2-norm	F bound	gap	$\frac{\|\Delta\mathbf{L}\|_2}{\text{gap}}$	$\|\sin\Theta\|_F$	true $\Delta\mathbf{L}$	chain	ARI
0.20	0.017	0.0078	0.0129	1.65	0.0111	0.141	0.0451	0.247	0.007	0.86	10.9	1.00
0.30	0.039	0.0164	0.0296	1.80	0.0234	0.324	0.0443	0.528	0.022	1.83	25.4	1.00
0.40	0.072	0.0268	0.0540	2.02	0.0381	0.592	0.0435	0.877	0.070	3.04	47.2	1.00
0.45	0.092	0.0322	0.0693	2.15	0.0459	0.759	0.0430	1.067	0.147	3.70	61.2	1.00
0.50	0.116	0.0377	0.0868	2.30	0.0537	0.951	0.0425	1.263	0.375	4.38	77.6	0.73
0.55	0.143	0.0431	0.1067	2.47	0.0614	1.169	0.0420	1.464	0.947	5.08	96.6	0.13
0.60	0.173	0.0484	0.1290	2.67	0.0689	1.413	0.0413	1.668	0.984	5.78	118.6	0.13
0.80	0.322	0.0673	0.2445	3.63	0.0957	2.678	0.0381	2.509	0.994	8.70	243.5	0.13
real	0.064	0.0156	0.0756	4.84	0.0212	1.053	0.0317	0.669	0.157	2.32	114.9	0.90

Table 6: Problem (1). "ratio" is the Step 1 bound over the actual movement; "2-norm" is the actual $\|\Delta\mathbf{L}\|_2$; "F bound" is $\sqrt{2p}$ times the Step 1 bound; "true $\Delta\mathbf{L}$" is the Davis-Kahan bound evaluated with the actual change; "chain" is (4). Any Davis-Kahan value above $\sqrt{k} = 1.732$ says nothing. ARI is that of the observed graph against the sectors; the clean graph scores 1.00 in every simulated row and 0.87 on the real data.

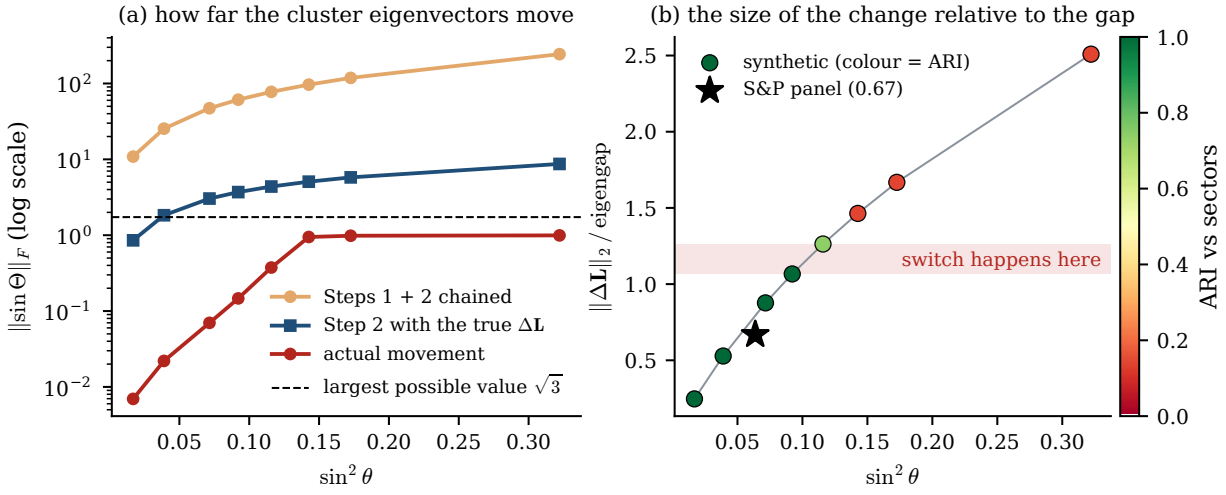


Figure 17: (a) How far the cluster eigenvectors actually move (red) against what the bound allows, with the true change in the graph (blue) and fully chained (orange). Everything above the dashed line is uninformative. (b) The change in the graph measured in units of the eigengap. The switch to beta clusters happens between 1.067 and 1.263 (shaded); the real data sits at 0.669.

- **Step 1 is valid and fairly tight**: the bound on the weight change exceeds the actual change by a factor 1.65 to 3.63 in simulation (2.15 to 2.47 near the switch) and 4.84 on the real data. *Reproduced.*
- **Passing to $\Delta\mathbf{L}$ loses a further factor of about 8.** At $\delta = 0.5$ the actual ratio $\|\Delta\mathbf{L}\|_F/\|\Delta\mathbf{w}\|$ is 1.43, against the worst case $\sqrt{2p} = 10.95$: the weight changes are spread over many links and largely cancel in each node's total.
- **Step 2 is uninformative here, even with the true $\Delta\mathbf{L}$.** The clean graph's eigengap is only about 0.04, so the Davis-Kahan bound passes $\sqrt{k}$ already at $\delta = 0.3$, where the subspace has in fact moved by 0.022. The full chain gives 10.9 to 243.5 against a ceiling of 1.732. *Proved,*

but vacuous on this problem.

- **The ratio $\|\Delta \mathbf{L}\|_2/\text{gap}$ does locate the switch.** At 0.877 and 1.067 nothing has switched ($\|\sin\Theta\|_F$ 0.070 and 0.147); at 1.263 one seed in three has (0.375); at 1.464 all have (0.947). The switch happens when the change reaches roughly the eigengap, which is the scale Davis-Kahan identifies; its constant and the slack in Step 1 put the certified threshold far below the real one.
- **On the real data the ratio is 0.669**, below every simulated row that switched. Unlike $\sin^2\theta$, this comparison does not depend on the sector strength, since the change and the gap are both measured on the data. It agrees with the finding that the built-in null suffices there. It is a measured rule on one design and one learner, not a theorem.

4.7 Part II conclusions

The Laplacian learner removes the market for free, exactly when betas are equal. When betas differ, what is left is a penalty on linking stocks with different betas. On the S&P data that penalty helps, because betas follow sectors, and explicit removal of the market makes clustering worse. When betas vary within sectors, the penalty can take over, and does so abruptly; explicit removal then recovers the sectors. The proved bound has the right form but constants far too loose to predict the switch; the measured rule "switch when the graph change reaches the eigengap" does.

5 Status, deviations and next steps

5.1 Status of every claim

claim	status	evidence
Part I		
(a) components counted by eigenvalue 1 of $\mathbf{B}$	proved; reproduced	Prop. 1; learned graph gives 1.0052, 1.0041 (degree tolerance)
(b) Fan term = relaxed modularity	proved; holds when $\lambda_k \leq 1$	exact form with $\max\{0, \lambda_k - 1\}$; two counterexamples; learned graph misses by 0.0093
(c) $\mu_1 = \max\{0, 1 - \lambda_2\}$	proved; unit degree only	does not apply to the time-varying graphs (Section 3.7)
regular / normalised / unequal degrees	proved	Table 1; $\mathbf{B}_n$ exact in every sweep run
k-means on $\mathbf{B}$ recovers the components	reproduced	ARI 1.000 on the learned graph and at every ρ
Part II		
$\mathrm{tr}(\mathbf{LS}) = \mathrm{tr}(\mathbf{LJSJ})$	proved; reproduced	graphs agree to 2.7×10^{-10} and 1.3×10^{-4}
market term is a beta-difference penalty, size $\propto \sin^2\theta$	proved	(3); unit tests
built-in null suffices on the S&P data	reproduced	Algorithm 1 ARI 1.000
explicit null improves on it (real data)	**not supported**	Algorithm 1: 0.830 to 0.896; Problem (1): 0.802 to 0.900
aligned unequal betas are harmless	reproduced (simulation)	ARI 1.000 in every aligned run
cross-cutting betas break the built-in null	reproduced (simulation)	switch at $\sin^2\theta$ 0.116 to 0.143 (Problem (1)), 0.143 to 0.173 (Algorithm 1)
explicit nulls fix that	reproduced (simulation)	ARI 1.000 in every run
Step 1 of the bound	proved; holds	within a factor 1.65 to 4.84
Davis-Kahan step	proved; **vacuous**	10.9 to 243.5 against a ceiling of 1.732
switch when $\|\Delta\mathbf{L}\|_2 \approx$ gap	observed, not proved	none at 1.067, partial at 1.263; real data 0.669

5.2 Deviations

- **Solver settings.** The Fan weight $\eta = 300$ is ours (the paper does not report it); the degree rule is a penalty with $\rho = 100$, so degrees are 1.0012 to 1.0069 rather than 1; Algorithm 1 starts from the Problem (1) graph, not the uniform graph. Problem (1) uses sparsity 10, the paper's Fig. 1 setting.
- **Market index.** `^GSPC` daily closes from Yahoo Finance; the paper uses no index.
- **Explicit nulls.** S-mode is rescaled to unit diagonal and RMT has its unit diagonal put back, so that both are correlation-like inputs.
- **Algorithm 1 did not reach three components** on three real inputs (S-mode, resid-SPX, RMT) and one simulated run (aligned, $\delta = 0.2$, seed 1, where $\lambda_2 = \lambda_3 = 0.0033$). Clusters there come from k-means on the bottom three eigenvectors; the simulated run still scores 1.000. None of the 135 fine-grid Algorithm 1 runs needed this.
- **The bound** is for Problem (1) only, as Algorithm 1 is not convex.

- **Simulation.** $\sin^2\theta$ uses population loadings in simulation and sample loadings on real data. The simulation's variances are our choice, and the switch location depends on them.
- **The ρ sweep** is ours, not the paper's.

5.3 Suggested next steps

Two directions seem worthwhile; which to take is the professor's decision.

- **Part III**, once a repair of its objective is chosen (18 October note). Parts I and II suggest what it should aim at: under unit degree modularity adds nothing, and on this data the built-in null already works, so any gain would have to come from letting degrees differ.
- **Part II on harder data.** The S&P panel is the easy case because betas follow sectors. A universe where betas vary strongly within sectors, for example a broader cross-section of the S&P 500, would test whether the simulated switch happens in practice, with $\|\Delta\mathbf{L}\|_2$/gap as the diagnostic.

5.4 Reproducing

From `modularity_graph_learning/`: `python report_numbers.py --part1` and `--part2` print every number in Sections 3 and 4 from the committed caches in seconds; `python report_figures.py` redraws every figure. The underlying runs are rebuilt by `degree_sweep.py`, `part2_runs.py` (about 45 minutes with two workers) and `part2_threshold.py`. Tests: `tests/test_lemma.py` (25) and `tests/test_part2.py` (9).

A Glossary

term	meaning
p, T	number of stocks (97 real, 60 simulated) and of trading days (753)
$\mathbf{S}$	correlation matrix of daily returns; the learners' only input
$\mathbf{W}$, W_{ij}	learned link weights between stocks
degree d_i, $\mathbf{d}$, $\mathbf{D}$	total link weight of stock i; the vector and diagonal matrix of degrees
$2m$	total weight, $\sum_i d_i$
unit degree, regular	every $d_i = 1$; every d_i equal to the same c
Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{W}$	measures how much a node signal changes across links, (1)
λ_i	eigenvalues of $\mathbf{L}$, increasing; $\lambda_1 = 0$
algebraic connectivity λ_2	0 exactly when the graph is disconnected; Paper 1's stress indicator
Fiedler vector	eigenvector of λ_2
eigengap	$\lambda_{k+1} - \lambda_k$: how clearly the graph has k groups
component	group of stocks linked among themselves and to no one else
null model	model of the weight expected by chance; here $d_i d_j / 2m$
modularity matrix $\mathbf{B}$	$\mathbf{W} - \mathbf{d}\mathbf{d}^\top / 2m$, observed minus expected weight
μ_i	eigenvalues of $\mathbf{B}$, decreasing
relaxed modularity	$\max \operatorname{tr}(\mathbf{U}^\top \mathbf{B} \mathbf{U})$ over orthonormal $\mathbf{U}$
divisibility test	a network is worth splitting only if $\mu_1 > 0$
$\mathbf{L}_{\text{sym}}$, $\mathbf{B}_n$	degree-normalised Laplacian and modularity matrix
Ky Fan's theorem	sum of the k smallest eigenvalues $= \min \operatorname{tr}(\mathbf{V}^\top \mathbf{L} \mathbf{V})$
gdet	generalised determinant: product of the nonzero eigenvalues
Problem (1)	Paper 1's basic learner (convex, connected output)
Algorithm 1	Paper 1's k-component, unit-degree learner
η, ρ, β	Fan weight (300), degree penalty (100), sparsity weight (10)
spectral clustering, k-means	clustering on the bottom eigenvectors
ARI	adjusted Rand index; 1 identical, about 0 chance
beta, loading b_i	sensitivity of stock i to the market; $\operatorname{corr}(x_i, f)$
$\mathbf{J}$	$\mathbf{I} - \mathbf{1}\mathbf{1}^\top / p$, removes the average across stocks
$\sin^2 \theta$	beta inequality: 0 when all betas are equal
S-mode, resid-SPX, resid-EW, RMT, oracle	explicit market removals (Section 4.2)
Marchenko-Pastur edge	$(1 + \sqrt{p/T})^2$, the largest eigenvalue pure noise produces
principal angles, $\|\sin \Theta\|_F$	distance between two subspaces; at most $\sqrt{k}$
strong convexity	curvature bounded below; minimisers move little
Davis-Kahan	eigenvectors move little if the change is small against the eigengap

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